tokens → concepts

VCM: Vision Concept Modeling

Adaptive Vision Token Compression via Instruction Fine-Tuning

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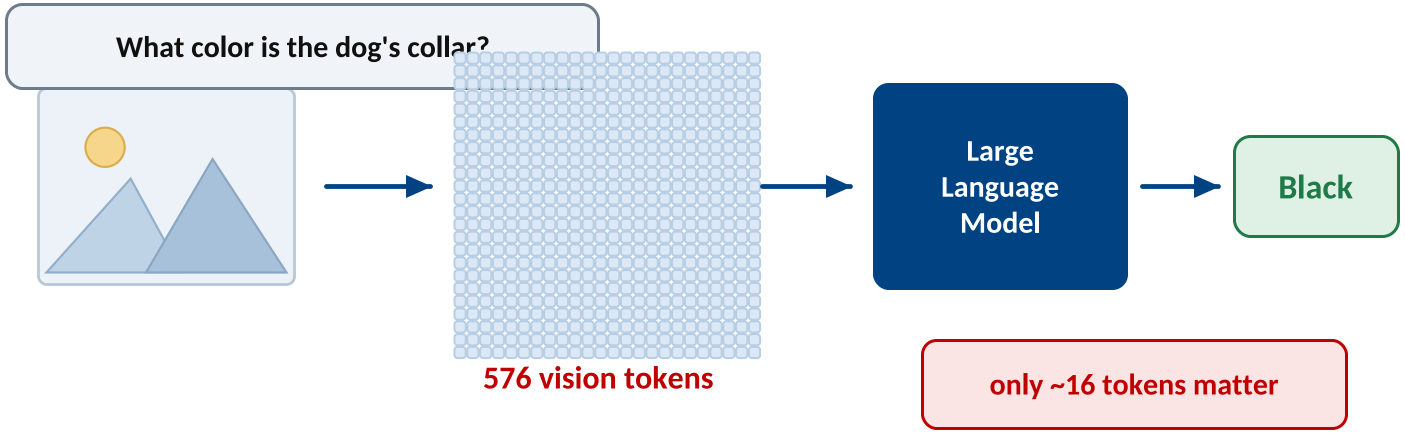
The bottleneck — LVLMs still read every image token

One simple question still activates every patch



Why the waste compounds

- **≈20×** vision vs text tokens
- **n^2** attention term
- **2,880** high-res tokens

Cost model

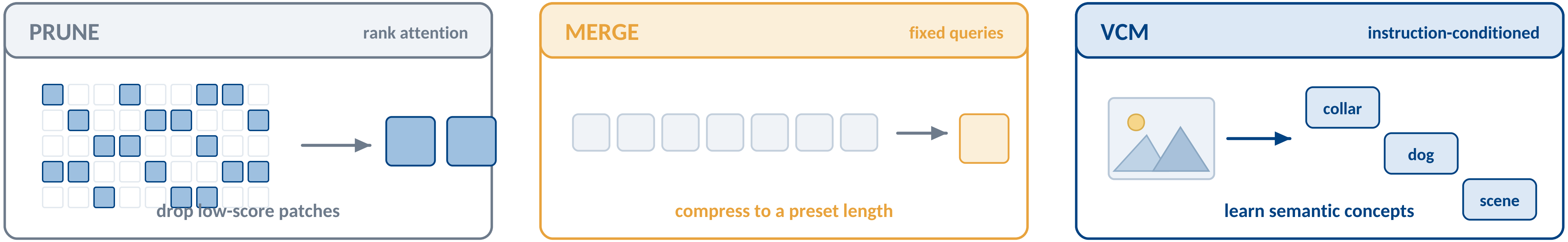
$$\text{FLOPs} = T \cdot (4nd^2 + 2n^2d + 2ndm)$$

reduce vision length → reduce the dominant term

The answer needs a concept; the model pays for the whole image.

Figure 1 and Appendix D of the paper.

Compression is not yet concept modeling



A vision concept model must satisfy all three

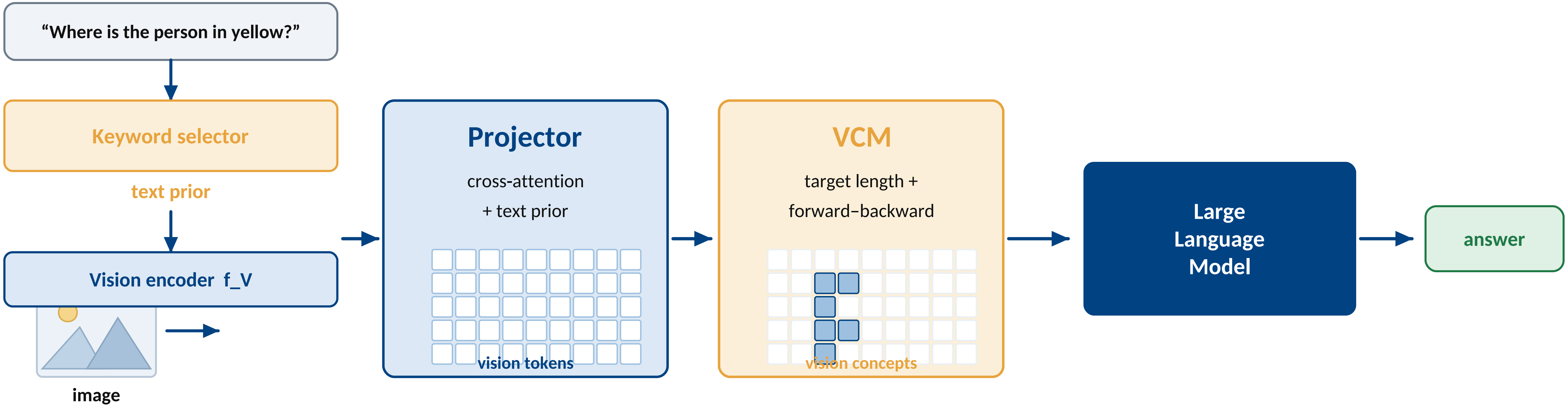
1. **HOW MANY** — adaptive quantity
2. **WHICH** — semantic identity
3. **WHERE** — spatial grounding

all conditioned on the instruction — not a fixed compression rule

A vision concept model must decide how many, which, and where — from the instruction.

Sections 1, 2, and 5.1 of the paper.

VCM — one model, two training stages



Stage I · pre-training

$L_{NTP} + \alpha \cdot L_{SA}$ α = 0.05 · learns the keyword selector

Stage II · instruction fine-tuning

$L_{NTP} + \varepsilon(r) \cdot L_{VCM}$ selector frozen · learns the concepts

Use the text prior to set a budget, then align tokens into concepts.

Section 3 and Figure 3(a) of the paper.

Self-supervision — the instruction tells us what vision is needed

1 SELECT
person · yellow

→

2 MASK
remove 0 → 2 keywords

→

3 MEASURE
instruction − response

→

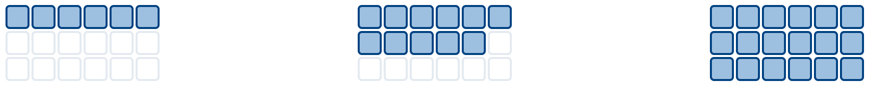
4 TARGET
adaptive length L

Semantic alignment exposes the useful words

Where	is	the	**person**	in	**yellow**	?

score > mean → keyword

Masking turns one example into a free contrast



0 mask → 6 · 1 mask → 11 · 2 mask → 18

$$L = \mathrm{floor}(\, M \cdot S \cdot [1 - \mathrm{Norm}(N_{key})]\,)$$

M input tokens · S information budget · Nkey keyword gap

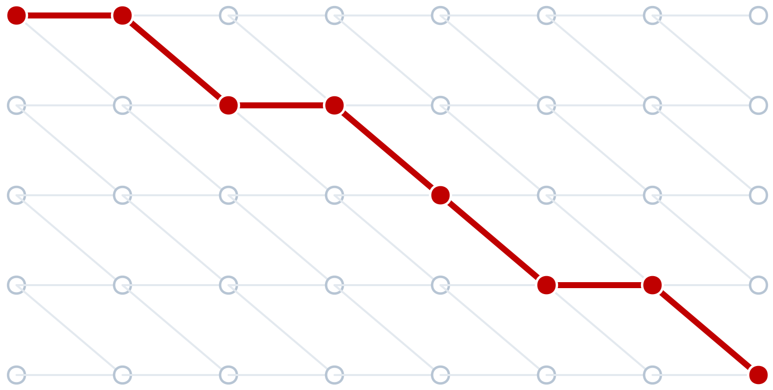
S=1/2	S=1/4	S=1/6	S=1/8
288 tok	**144 tok**	72 tok	36 tok

Table 4 sweet spot

Keywords provide the signal; masking creates the contrast; no concept labels are required.

Optimization — learn variable-length concepts without choosing a cut

1 · Sum over every monotone alignment



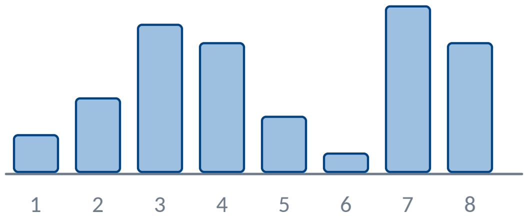
one path shown · all paths contribute

2^M enumeration → dynamic programming M^2

2 · Learn the cut

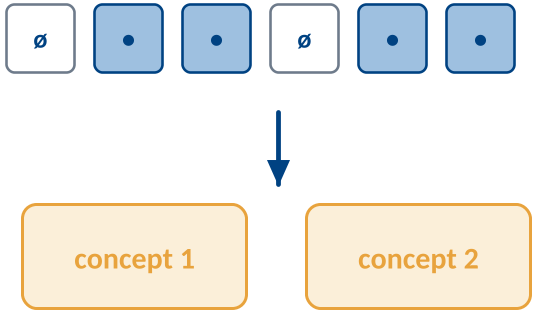
$$\partial L/\partial a = p - \gamma/Z$$

prediction posterior



posterior residual · CTC-shaped gradient

3 · Merge kept runs



score-weighted average

parallel implementation · 100×

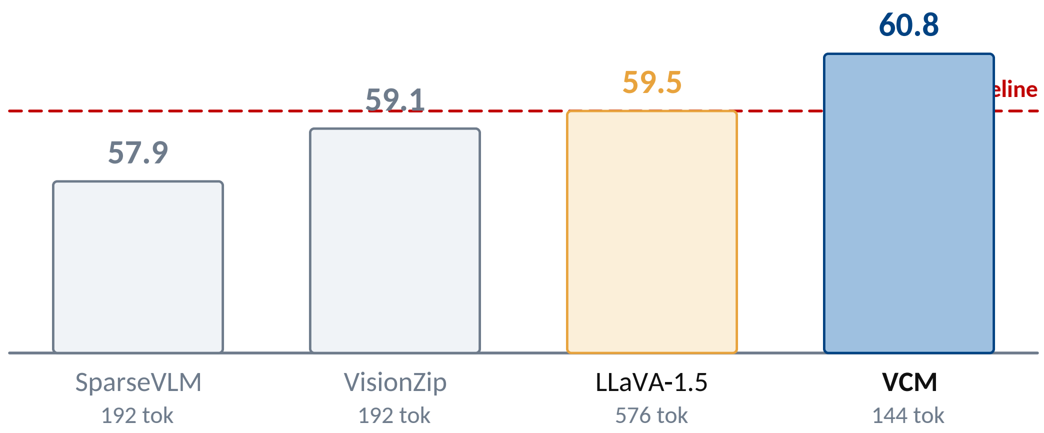
Unknown boundaries are marginalized during training — not guessed in advance

Forward–backward sums every alignment, then adjacent kept tokens merge into concepts.

Sections 3.3–3.4, Appendix C, Table 6, and Algorithm 2.

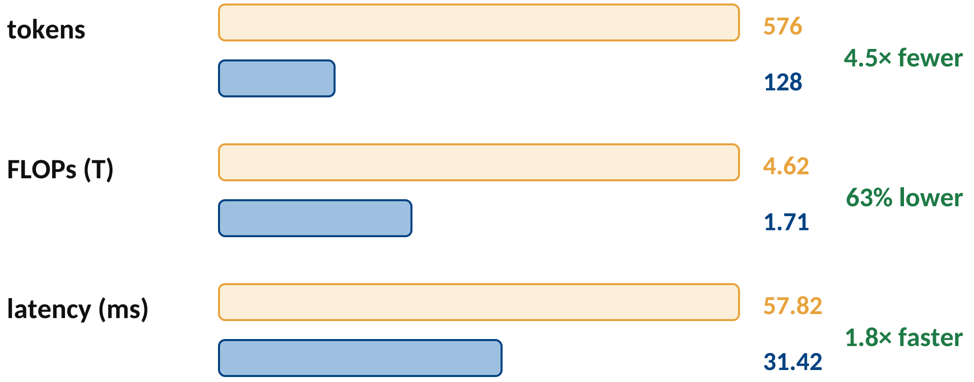
Core evidence — less computation, no performance trade-off

11-benchmark VQA average



Avg. (%) — vertical axis starts at 54.0

Measured efficiency · Table 10



144 vision tokens
vs 576 baseline

60.8 VQA average
vs 59.5 baseline

85% fewer FLOPs
analytic at S = 1/4

+13.6 COCO stuff Top-1
11.8 → 25.4

At 144 tokens VCM beats the 576-token baseline; at S = 1/4 it uses 85% fewer FLOPs.

Tables 1 and 10; Appendix D.

What VCM changes — efficiency, representation, and scope

HIGH RES
2,880 → 160 tok
70.1 avg

VIDEO
2,048 → 136 tok
52.5 avg

QWEN2-VL
1,326 → 576 tok
69.6 avg

DENSE
COCO · ADE
all improve

Three contributions

1. **Definition** — quantity · identity · location
2. **Learning** — text prior + forward–backward
3. **Outcome** — efficiency + dense perception

What remains unresolved

Keyword bias
selected words may not be the true concepts

Coarse mapping
min–max length normalization is simplified

The transferable idea: let the task decide the visual granularity

Model the concepts the instruction needs — not every token the image has.

Run Luo, Renke Shan, Longze Chen, Ziqiang Liu, Lu Wang, Min Yang, Xiaobo Xia. VCM: Vision Concept Modeling with Adaptive Vision Token Compression via Instruction Fine-Tuning. arXiv:2504.19627v2, 2025.